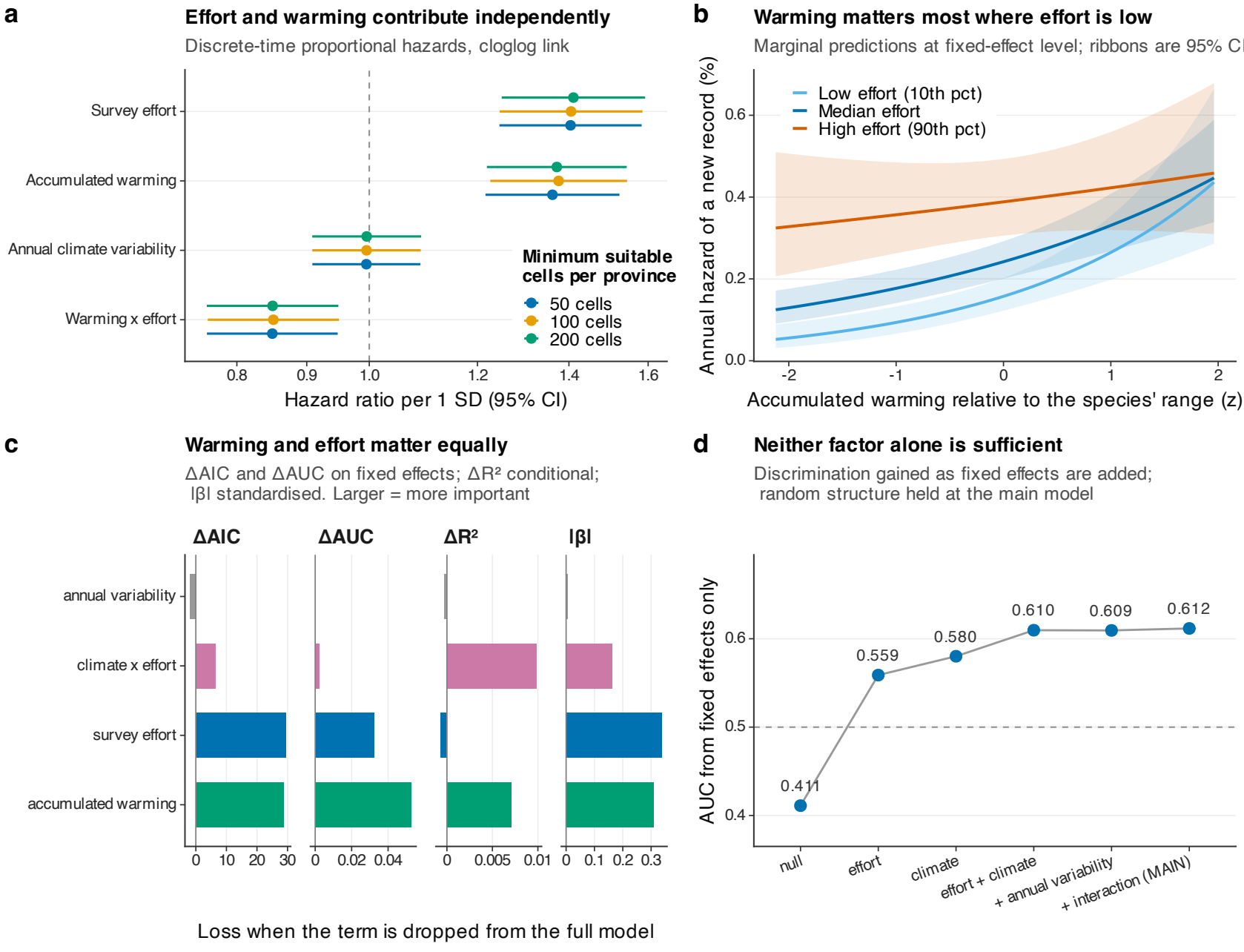
a
Effort and warming contribute independently
Discrete-time proportional hazards, cloglog link
Survey effort
Accumulated warming
Annual climate variability
Warming x effort
Minimum suitable cells per province
50 cells
100 cells
200 cells
0.8
0.9
1.0
1.2
1.4
1.6
Hazard ratio per 1 SD (95% CI)
b
Warming matters most where effort is low
Marginal predictions at fixed-effect level; ribbons are 95% CI
Low effort (10th pct)
Median effort
High effort (90th pct)
Annual hazard of a new record (%)
0.0
0.2
0.4
0.6
-2
-1
0
1
2
Accumulated warming relative to the species' range (z)
c
Warming and effort matter equally
ΔAIC and ΔAUC on fixed effects; ΔR² conditional;
|β| standardised. Larger = more important
ΔAIC
ΔAUC
ΔR²
|β|
annual variability
climate x effort
survey effort
accumulated warming
0
10
20
30
0
0.02
0.04
0
0.005
0.01
0
0.1
0.2
0.3
Loss when the term is dropped from the full model
d
Neither factor alone is sufficient
Discrimination gained as fixed effects are added;
random structure held at the main model
AUC from fixed effects only
0.4
0.5
0.6
0.411
0.559
0.580
0.610
0.609
0.612
null
effort
climate
effort + climate
+ annual variability
+ interaction (MAIN)